Ex – 4 Solns

1. Create an array "fruit" which has 5 objects. Each object has properties called "title" and "colour". Write a function to add a new fruit without mutating the existing array.

```
<!DOCTYPE html>
<html>
<head>
   <title>Add Fruit</title>
</head>
<body>

<h2>Fruit Details</h2>

<label>Fruit Name:</label>
<input type="text" id="title"><br><br>
```

```
<label>Colour:</label>
<input type="text" id="colour"><br><br>

<button onclick="addFruit()">Add
Fruit</button>

<h3>Updated Fruit List</h3>
<p id="output"></p>

<script>

const fruit = [
    { title: "Apple", colour: "Red" },
    { title: "Banana", colour: "Yellow" },
    { title: "Orange", colour: "Orange" },
    { title: "Mango", colour: "Yellow" },
    { title: "Grapes", colour: "Green" }
];

function addFruit() {
```

```
    let title =
document.getElementById("title").value;
    let colour =
document.getElementById("colour").value;


    let newFruit = { title, colour };

    let newArray = [...fruit, newFruit];

    let result = "";

    for (let i = 0; i < newArray.length; i++) {

        let item = newArray[i];

        result = result + item.title + " - " +
item.colour + "<br>";
    }
```

```
document.getElementById("output").innerH
TML = result;
}

</script>

</body>
</html>
```

2. Create an array of Vegetables. Write a function to print only the names of vegetables whose length is more than 5. (Hint: use Array.filter())

```
<!DOCTYPE html>
<html>
<head>
    <title>Filter Vegetables</title>
</head>
<body>
```

```
<h2>Vegetables with More Than 5 Letters</h2>

<button onclick="showVegetables()">Show Vegetables</button>

<p id="output"></p>

<script>

const vegetables = [
    "Potato",
    "Carrot",
    "Onion",
    "Tomato",
    "Radish",
    "Peas",
    "Cabbage"
];

function showVegetables() {
```

```
    // Filter vegetables whose length is greater
than 5
    let result =
vegetables.filter(function(item) {
        return item.length > 5;
    });


document.getElementById("output").innerH
TML = result.join("<br>");
}


</script>

</body>
</html>
```

3. Create an array of 5 random words. Write a function to find the longest

spelled word in the array. (Hint: use Array.reduce())

```
<!DOCTYPE html>
<html>
<head>
    <title>Longest Word</title>
</head>
<body>

<h2>Find the Longest Word</h2>

Word 1: <input type="text"
id="word1"><br><br>
Word 2: <input type="text"
id="word2"><br><br>
Word 3: <input type="text"
id="word3"><br><br>
Word 4: <input type="text"
id="word4"><br><br>
```

```
Word 5: <input type="text"
id="word5"><br><br>

<button onclick="findLongest()">Find
Longest Word</button>

<p id="output"></p>

<script>

function findLongest() {

    let words = [

document.getElementById("word1").value,

document.getElementById("word2").value,

document.getElementById("word3").value,

document.getElementById("word4").value,
```

```
document.getElementById("word5").value
    ];

    let longest = words.reduce(function(a, b)
{

        if (a.length > b.length) {
            return a;
        }
        else {
            return b;
        }

    });


document.getElementById("output").innerH
TML =
    "Longest Word: " + longest;
```

```
}

</script>

</body>
</html>
```

4. Create an array of 5 numbers by getting input from the user. Write a pure function, which allows the user to replace a number in the array with a new number.

```
<!DOCTYPE html>
<html>
<head>
    <title>Replace Number using
map()</title>
</head>
<body>
```

```
<h2>Replace a Number</h2>

Number 1: <input type="number"
id="n1"><br><br>
Number 2: <input type="number"
id="n2"><br><br>
Number 3: <input type="number"
id="n3"><br><br>
Number 4: <input type="number"
id="n4"><br><br>
Number 5: <input type="number"
id="n5"><br><br>

Number to Replace:
<input type="number"
id="oldNumber"><br><br>

New Number:
<input type="number"
id="newNumber"><br><br>
```

```
<button
onclick="replaceNumber()">Replace</butto
n>

<p id="output"></p>

<script>

function replaceNumber() {

    let numbers = [

Number(document.getElementById("n1").v
alue),

Number(document.getElementById("n2").v
alue),

Number(document.getElementById("n3").v
alue),
```

```
Number(document.getElementById("n4").value),

Number(document.getElementById("n5").value)
    ];

    let oldNum = Number(document.getElementById("oldNumber").value);
    let newNum = Number(document.getElementById("newNumber").value);

    let newArray = numbers.map(function(item) {

        if (item == oldNum) {
            return newNum;
        }
```

```
        else {
            return item;
        }

    });


document.getElementById("output").innerHTML =
    "New Array: " + newArray;

}

</script>

</body>
</html>
```

5. Working with Array Methods:

```
students = [
  { name: "Alice", score: 85 },
```

```
  { name: "Bob", score: 58 },
  { name: "Charlie", score: 95 },
  { name: "David", score: 42 },
  { name: "Eva", score: 75 }
]
```

a. Display the names of the students who have passed the exam with score above 60.
b. Find the average score of this class.
c. Modify the score as percentage (85%).

```
<!DOCTYPE html>
<html>
<head>
    <title>Array Methods</title>
</head>
<body>

<button onclick="showResult()">Show Result</button>
```

```
<p id="output"></p>

<script>

const students = [
    {name:"Alice", score:85},
    {name:"Bob", score:58},
    {name:"Charlie", score:95},
    {name:"David", score:42},
    {name:"Eva", score:75}
];

function showResult(){

    // a. Students who passed
    let passed =
students.filter(function(student){
        return student.score > 60;
    });

    // b. Average score
```

```
    let total = students.reduce(function(sum, student){
        return sum + student.score;
    },0);

    let average = total / students.length;

    // c. Score as percentage
    let percent = students.map(function(student){
        return student.score + "%";
    });


document.getElementById("output").innerHTML =
    "Passed Students: " + passed[0].name + ", " + passed[1].name + ", " + passed[2].name +
    "<br><br>Average Score: " + average +
    "<br><br>Percentage: " + percent;
```

```
}

</script>

</body>
</html>
```